



OPERATOR GUIDE

Box3D Metal Quick Start

Reconstruct the pinned Box3D checkout, apply the Metal overlay, validate it, and enable GPU routing per world.



Reconstruct and run

The public repository is an overlay: it contains no Box3D checkout. The bootstrap script clones the canonical upstream revision, verifies the patch, applies it, and builds a Release configuration.

Requirements

- Apple Silicon Mac and macOS Metal
- Xcode command-line tools
- Git, CMake, and Ninja

```
git clone https://github.com/Lulzx/box3d-metal.git
cd box3d-metal
./scripts/bootstrap.sh ../box3d-metal-worktree
```

Validation

```
../box3d-metal-worktree/build/metal-release/bin/test MetalTest
../box3d-metal-worktree/build/metal-release/bin/metal_demo
```

For a disposable end-to-end reconstruction, run `./scripts/verify-clean.sh`.

Enable per world

```
b3WorldId world = b3CreateWorld(&worldDef);
if (!b3World_EnableMetal(world, 32768)) {
/* CPU path remains usable. */
}
```

The body threshold is deliberately caller-selected. It is not a universal recommendation; measure your world with full-step timing.

Read route telemetry

```
b3MetalProfile p = b3World_GetMetalProfile(world);
printf("%s contacts=%llu joints=%llu fallbacks=%llu\n",
p.deviceName, p.contactDispatchCount,
p.jointDispatchCount, p.contactFallbackCount);
```

Dispatch counters prove a Metal stage ran. They do not imply that broad phase, narrow phase, CCD, sleeping, events, or finalization ran on the GPU.

Next references

- docs/compatibility.md - exact supported and CPU fallback surface
- docs/performance.md - measured M4 Pro crossovers
- docs/troubleshooting.md - initialization, routing, and performance diagnosis